

Wrist Band GSR PPG IMU MAX30009 MAX30102 BNO055

Parts Selection

Component	Description	
BQ25100YFPR	Battery Charger from USB-C. Very low power consumption when idle to Increase battery Life.	
TLV75533PDBVR	3.3V LDO for ESP32-C3	
TCR2EF18_LMCT	1.8V LDO for other low power components	
MAX30102EFD+T	PPG Module for high-sensitivity Pulse Oximeter and Heart-Rate sensor	
MAX30009ENA+T	Most Reliable and highly configurable GSR Module for exclusive GSR measurement and experiment.	
BNO055	IMU	

All the peripheral are configured for communicating over I2C with ESP32-C3 .

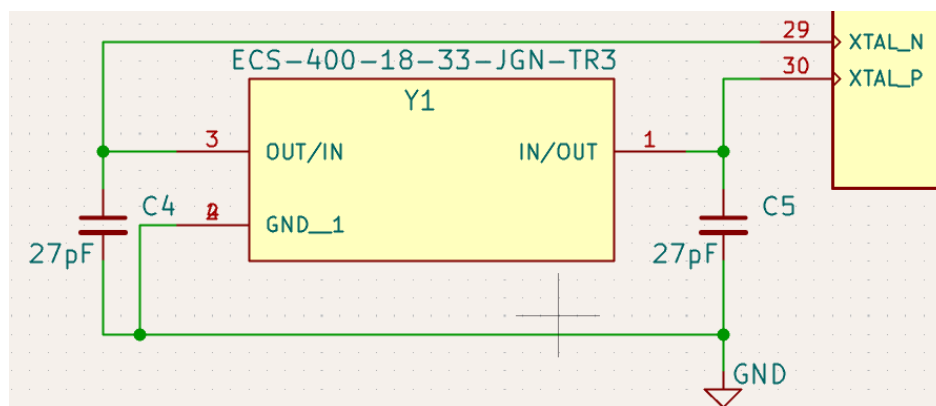
Crystal Selection and load capacitance

Crystal Model : ECS-400-18-33-JGN-TR3

Load Capacitance is 18pF.

ESP32-C3's Pin 29 and Pin 30 Load capacitance 1pF to 2 pF

Board Stray Capacitance 2 to 3 pF



Chosen capacitor for crystal 27pF. So it matches 18pF Load capacitance.

$$C_L = \frac{27 \times 27}{27 + 27} + C_{stray} = 13.5 \text{ pF} + C_{stray}$$

Typical stray capacitance sources:

Source	Value
PCB trace parasitics	~1-2 pF
ESP32-C3 XTAL pin capacitance	~2-3 pF
Total C_stray	~3-5 pF

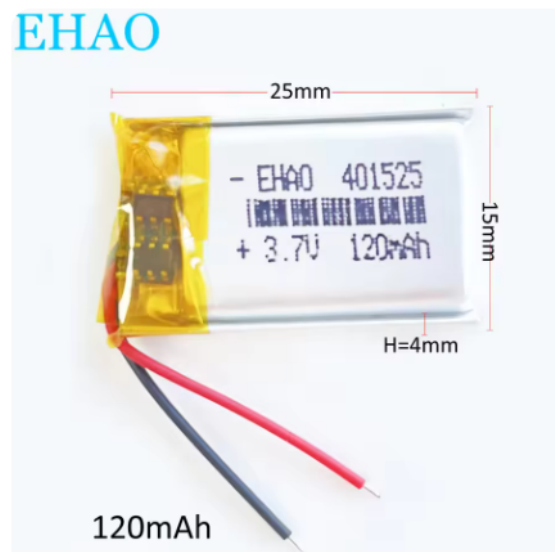
$$C_L \approx 13.5 + 4 \approx 17.5 \text{ to } 18.5 \text{ pF}$$

Battery Selection

Battery Sourcing: **3.7V 120mAh Lipo Rechargeable Battery 401525 For MP3 GPS Headphone Headset Camera DVD Bluetooth Smart Watch**

[https://ko.aliexpress.com/item/1005009816163946.html?](https://ko.aliexpress.com/item/1005009816163946.html?spm=a2g0o.cart.0.0.1db938daVRcWRI&mp=1&pdp_npi=6%40dis%21KRW%21KRW+4635%21KRW+3940%21%21KRW+394)

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PCBWAY HDI PCB Capabilities

Generally better quality wrist band style PCB is not possible without HDI PCB. Here is the PCBWAY HDI PCB Capabilities:

Feature	Capability
Quality Grade	Standard IPC 2
Number of Layers	4 - 24layers
Order Quantity	1pc - 10000+pcs
Build Time	2days - 5weeks
Material	FR4 standard Tg 140°C,FR4 High Tg 170°C, FR4 and Rogers combined lamination
Board Size	Min 6*6mm Max 457*610mm
Board Thickness	0.4mm - 3.0mm
Copper Weight (Finished)	0.5oz - 2.0oz
Min Tracing/Spacing	2.5mil/2.5mil
Solder Mask Sides	As per the file
Solder Mask Color	Green, White, Blue, Black, Red, Yellow
Silkscreen Sides	As per the file
Silkscreen Color	White, Black, Yellow
Surface Finish	HASL - Hot Air Solder Leveling Lead Free HASL - RoHS ENIG - Electroless Nickle/Immersion Gold - RoHS Immersion Silver - RoHS Immersion Tin - RoHS OSP - Organic Solderability Preservatives - RoHS
Min Annular Ring	4mil, 3mil - laser drill
Min Drilling Hole Diameter	6mil, 4mil - laser drill
Max Exponents of Blind/Buried Vias	stacked vias for 3 layers interconnected, staggered vias for 4 layers interconnected
Other Techniques	Flex-rigid combination Via In Pad Buried Capacitor (only for Prototype PCB total area $\leq 1\text{m}^2$)